

Big Data in the Big Apple: Citi Bike Trips Analysis based on Weather Data

Project Overview:

This project investigates the correlation between hourly weather conditions and Citi Bike ridership patterns in New York City during September 2025. By integrating high-volume trip data with granular meteorological observations, the study aims to identify the specific "tipping points" where weather factors (such as precipitation thresholds and "feels like" temperatures) cause significant shifts in transit behavior across different user demographics.

Data Sources:

- 1) Citi Bike System Data - A comprehensive dataset consisting of over **5.2 million trip records** for September 2025. Key variables include start/end timestamps, station geodata, rideable type (Classic vs. Electric), and user type (Member vs. Casual). ([link](#))
- 2) Visual Crossing Weather API - Hourly historical weather data for New York City. This provides localized metrics including **temperature, feelslike values, precipitation levels, wind speed**, and general atmospheric conditions. ([link](#))

Questions to Answer:

- 1) How does **rain** and **feelslike temperature** affect Citi bike rides?
- 2) How does ridership volume compare between **"Overcast" conditions with zero precipitation versus "Clear" sunny days?**
- 3) Does decision to use bike vary if you are **not a member?**